

employed in each ISCO-08 1-digit group and from expert opinions. As the final step, we provide updated global, regional and income-based estimates of employment that has the potential to be transformed by Gen AI technology.

Our study specifically targets GenAI tools, isolating the effects of this recent family of AI from broader AI applications, such as the use of Machine Learning (ML) algorithms or image recognition in industrial production. There are several considerations that justify such selective focus. First, we respond to the growing need for a research tool that could provide a better picture of GenAI's potential impact on national labour markets, given the growing societal angst concerning this technology and the recent reports about increasing levels of adoption of GenAI at work (Bick et al., 2024; Faverio and Tyson, 2022; Grampp et al., 2023; Maison & Partners and ThinkTank, 2024; Pew Research Center, 2023; Rutgers, 2024). Second, while several indicators of occupational exposure to broader AI technologies are available to researchers (Nurski and Vansteenkiste, 2024), very few tools allow for a more detailed focus on GenAI (Nurski and Ruer, 2024). A new, more precise exposure index enables a closer alignment of the academic work on digital economy and labour with the significant interest in the GenAI technology in the media and in the public debate.¹ Third, the methodological blueprint provided by our research can be quite flexibly expanded to other types of AI in the future or focused more narrowly on specific subsets of digital technologies and sectors of their application.

This paper is the first in a series of upcoming publications based on ILO and NASK cooperation that strive to improve the methodology for assessing employment effects of AI, but also to document in detail the methods used, including the survey, expert verification process and the construction of the AI model used for predictions. Subsequent papers will present an assessment of the potential impact of GenAI on the Polish labour market, based on more detailed occupation and task-level data and using a mix of quantitative and qualitative methods. All publications will include the methodological blueprints and technical details of both the quantitative and qualitative components, including survey questionnaires and interview guidelines. By making these tools openly accessible, we hope to contribute to the development of an improved method for assessing potential impact of GenAI on jobs and to stimulate more detailed national-level studies, including in lower- and middle-income countries.

¹ For example, Felten et al. (2023) provided an updated version of their original exposure index from 2021, adjusting for the abilities of GenAI tools. As shown in their paper, the adjustment does not result in a major shift in occupations exposure and still makes it impossible to assess the abilities of this new technology in a selective manner. See Gmyrek et al. (2024) for a more detailed comparison of these scores.

► 1 Task-based approaches to automation in the literature

Central to the study of technology on work is the insight that jobs are a “bundle of tasks” (Autor, 2015). As such, task automation might, or might not lead to job automation, depending on the importance of a particular task to an occupation. Analyses of employment effects of automation that use the task-based approach, including this study, attempt to capture two central considerations: (i) the ability of a specific technology to automate a given task and (ii) the character of existing occupations in the labour market, based on the detailed tasks performed by these occupations. By modelling the intersection of these two dimensions, one arrives at a range of possible projections of how such technology could interact with occupational tasks, leading either to their full automation through technology-driven substitution, or to a transformation, through partial automation and human-machine complementarity. While the task-based framework certainly has limitations, recent academic literature shows the analytical advantages of this method for modelling employment and labour market impacts (Acemoglu et al., 2024; Nurski and Ruer, 2024).

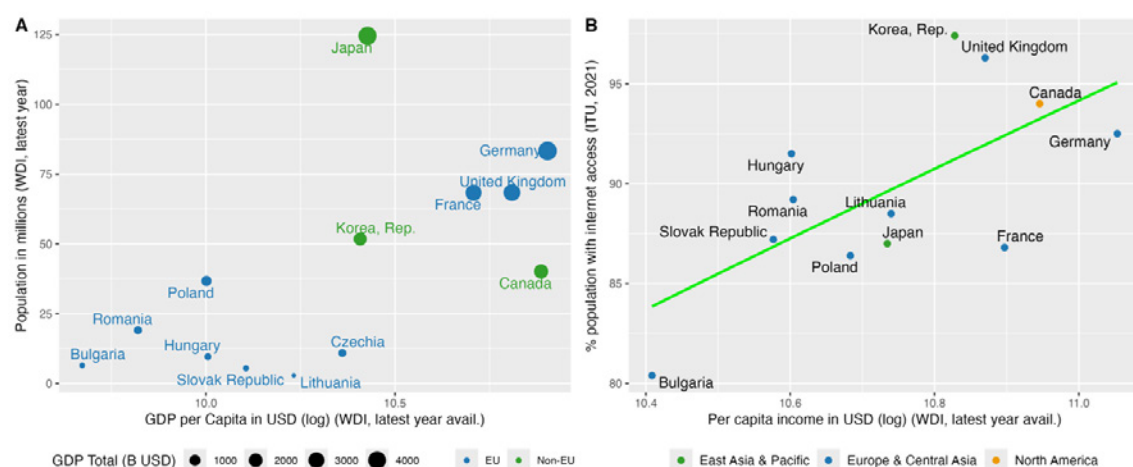
Research on AI and tasks has shown important improvements in methods and scope in recent years, which includes using AI tools for research purposes. In one of the early papers on Machine Learning (ML) and jobs, Frey and Osborne, (2013) fed the opinions of a group of IT technology experts into an ML algorithm to develop occupational-level projections of automation potential for the US job market. The emergence of GenAI APIs has simplified the possibility of combining human expertise and algorithmic capacities for research purposes. Relying on this idea, Eloundou et al. (2023) demonstrated a close alignment of GPT-4 predictions with a survey of 70 AI experts on the potential of automating occupational tasks with LLMs, and subsequently built on this feature to develop synthetic automation scores for the US market.

Most task-based studies are focussed on the US, due to the very detailed nature of the O*NET database, as well as its public accessibility. This typically forces researchers interested in the global picture or other countries and regions to base their analysis on the strong assumption of similarity of local economic and labour effects to those projected in the US context. In reality, occupational tasks can vary significantly among individual countries, which poses a particular challenge in developing countries. As such, working with tasks in the ISCO-08 system offers a useful and simple common denominator for projections across different countries, regions and income groups. Our method combines the strengths of this global framework with a significantly more detailed national classification system in Poland, which provides a detailed list of occupations and tasks, based on the latest labour market data. This helps us establish a starting point for modelling interaction of GenAI’s abilities with up-to-date tasks and occupations in a setting located between the countries with highest incomes and the emerging economies (Figure 1). Using a combination of GenAI tools, we subsequently transfer this knowledge back to the tasks in ISCO-08 system.

► 2 Tasks and occupations in the 6-digit system in Poland

Within Europe, Poland is quite representative of the larger group of Central and Eastern European (CEE) countries, albeit with a significantly larger population (38 million) than most of its peers. Globally, the CEE ranks between the high-income advanced economies and the emerging economies of the Global South. With respect to access to the internet, Poland ranks below UK and Germany, but higher than Japan and France. Thus, for assessing the potential of task automation, it seems reasonable to assume that Poland is representative of the upper threshold of automation potential, due to high availability of digital equipment and widespread internet access.

► Figure 1. Income- and population-based similarities (A) and access to the internet (B) across countries



The Ministry of Family, Labour and Social Policy (MOL) in Poland maintains a system of occupational classifications, with task and job descriptions that are regularly updated.² The system is used by Statistics Poland (GUS) and other state institutions for labor market analysis. At the most detailed, 6-digit level, this framework contains 2541 occupations and 29,753 corresponding tasks (Table 1). Since the system is publicly available online, we initially designed a scraping bot that crawled through individual occupational pages and downloaded job descriptions and corresponding tasks into a structured data frame.³ The extracted content was subsequently cross verified against a tabular occupational classification provided by the Ministry. The Polish classification system is compatible with the hierarchical structure of ISCO-08, which means that we could easily append an additional mapping of occupations from 1 to 4-digit level (Table 1).

² The occupational classification database (KZIS) was first published in 2014 and has been regularly updated, with the latest validation on 1 January 2025. The classification was introduced by the Regulation of the Minister of Labour and Social Policy on the classification of occupations and specializations for labor market purposes and its scope of application, dated 7 August 2014 (Dz.U. z 2018 r. poz. 227, z 2021 r. poz. 2285, z 2022 r. poz. 853, Dz.U. z 2024 r. poz. 1372). Additionally, the Ministry publishes a mapping to ISCO-08. See: <https://psz.praca.gov.pl/rynek-pracy/bazy-danych/klasyfikacja-zawodow-i-specjalnosci>.

³ Web scraping was primarily conducted to demonstrate the potential of acquiring non-traditional data from publicly accessible sources, in contexts where such activity is not restricted by server settings or legal regulations. The process adhered to responsible data collection practices, including appropriate frequency settings to avoid overloading servers.

► Table 1. Composition of occupational tasks in the Polish classification system and in ISCO-08

ISCO-08 occupational classification										Polish 6-digit occupational classification				
ISCO-08 1-digit	Group Name	Count of 2-digits	Share in 2-digits	Count of 3-digits	Share in 3-digits	Count of 4-digits	Share in 4-digits	Count of 6-digits	Share in 6-digits	Count of 6-digits	Share in 6-digits	Count of 6-digits	Share in 6-digits	Share in 6-digits
1	Managers	4	10.0%	11	8.7%	30	7.1%	173	6.8%	2565	8.6%			
2	Professionals	6	15.0%	27	21.4%	92	21.7%	721	28.4%	8715	29.3%			
3	Technicians and associate professionals	5	12.5%	19	15.1%	80	18.9%	507	20.0%	5956	20.0%			
4	Clerical support workers	4	10.0%	8	6.3%	27	6.4%	71	2.8%	768	2.6%			
5	Service and sales workers	4	10.0%	13	10.3%	40	9.5%	149	5.9%	1730	5.8%			
6	Skilled agricultural, forestry and fishery workers	3	7.5%	9	7.1%	17	4.0%	53	2.1%	684	2.3%			
7	Craft and related trades workers	5	12.5%	14	11.1%	65	15.4%	407	16.0%	4554	15.3%			
8	Plant and machine operators, and assemblers	3	7.5%	14	11.1%	40	9.5%	348	13.7%	3769	12.7%			
9	Elementary occupations	6	15.0%	11	8.7%	32	7.6%	112	4.4%	1012	3.4%			
	Total	40	100%	126	100%	423	100%	2541	100%	29753	100%			

Note: Table 1 illustrates the hierarchical relationship between ISCO-08 and the Polish 6-digit classification of occupations. The "Count of x-digits" columns show the number of occupations at each level, with 423 occupations at the 4-digit ISCO-08 level. The "Count of 6-digits" column represents the number of occupations in the Polish system mapped to ISCO-08's 4-digit level. Armed Forces Occupations have been excluded due to the lack of specific tasks. Shares indicate the percentage of each group at the corresponding level. The table only includes occupations for which detailed tasks are available in the Polish classification; those without tasks have been excluded.